

Bocconi

COMPUTER SCIENCE

code 30424 a.y. 2024-2025

Lesson 4

Using logical values in functions



Università
Bocconi
MILANO

Objectives of the lesson

Learn how to use logical values and logical functions



Our first case

In the **Sales** worksheet there's a list of apartments for sale in different areas of a big city.
We need to:

- Use a function in a new column named **Apartment size** that shows, for each apartment, the text "Big Size" if the apartment area is greater than or equal to 110 sqm, otherwise a blank cell
- Calculate, in a new column named **Final price**, the final selling price of each apartment, considering that when the apartment has a private garage and it is in excellent condition, the owner's selling price must be increased by 50,000 euros
- Calculate, in a new column named **Sales commission**, the commission to be paid for each apartment, considering that the commission is 5% for apartments in the areas with code C02 and A06, 3% in all other areas

Our first case

- Use a function in a new column named **SQM Analysis** that shows, for each apartment and using the Final price column values:
 - The text “Cheap sqm price” if the square meter price is less than 4,200 euros
 - The text “Normal sqm price” if the square meter price is between 4,200 and 6,000 euros (both inclusive)
 - The text “Expensive sqm price” if the square meter price is greater than 6,000 euros
- In a new column named **Interesting apartments**, use a function to show, for each apartment, the text “Big deal” when the apartment condition is normal or excellent and the sqm price is less than 5,000 euros (use values of the Final price column)

Our second case

In cell B6 of the **Grade converter** worksheet we need to use a function to convert a score (1-100) into an American grade (A-F), as shown in the table below:

Score	Grade
91-100	A
81-90	B
71-80	C
61-70	D
0-60	F



Tools to learn with today's lecture

- Logical functions
- IF function
- AND, OR functions
- Nested IF functions
- IFS function

Book references

Excel for students in economics and finance, 2nd edition

Chapters 4.1.1, 4.1.2, 4.1.3, 4.1.4, 4.1.5

Excel workbook, 4th edition:

Exercises on the functions and tools of the lesson

